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*Original Article*

# Artificial intelligence in a prediction model for postendoscopic retrograde cholangiopancreatography pancreatitis

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***Objectives:*** In this study we aimed to develop an artificial intelligence-based model for predicting postendoscopic retrograde cholangiopancreatography (ERCP) pancreatitis (PEP).

***Methods:*** We retrospectively reviewed ERCP patients at Nagoya University Hospital (NUH) and Toyota Memorial Hospital (TMH). We constructed two prediction models, a random forest (RF), one of the machine-learning algorithms, and a logistic regression (LR) model. First, we selected features of each model from 40 possible features. Then the models were trained and validated using three fold cross-validation in the NUH cohort and tested in the TMH cohort. The area under the receiver operating characteristic curve (AUROC) was used to assess model performance. Finally, using the output parameters of the RF model, we classified the patients into low-, medium-, and high-risk groups.

***Results:*** A total of 615 patients at NUH and 544 patients at TMH were enrolled. Ten features were selected for the RF model, including albumin, creatinine, biliary tract cancer, pancreatic cancer, bile duct stone, total procedure time, pancreatic duct injection, pancreatic guidewire-assisted technique without a pancreatic stent, intraductal ultrasonography, and bile duct biopsy. In the three fold cross-validation, the RF model showed better predictive ability than the LR model (AUROC 0.821 vs. 0.660). In the test, the RF model also showed better performance (AUROC 0.770 vs. 0.663, $P = 0.002$). Based on the RF model, we classified the patients according to the incidence of PEP (2.9%, 10.0%, and 23.9%).

***Conclusion:*** We developed an RF model. Machine-learning algorithms could be powerful tools to develop accurate prediction models.

**Key words:** artificial intelligence, endoscopic retrograde cholangiopancreatography, machine learning, pancreatitis, risk factor

## INTRODUCTION

ENDOSCOPIC RETROGRADE CHOLANGIOPANCREATOGRAPHY (ERCP) is widely used to diagnose and treat pancreatobiliary diseases. However, complications due to this procedure can occur, and the frequencies have been reported as follows: bleeding (0.3–2%), cholangitis (0.5–3%), cholecystitis (0.5–5.2%), and perforations (0.08–0.6%). In addition, an important complication is post-ERCP pancreatitis (PEP), which reportedly occurs at a frequency of 3–10% and has a mortality rate of 0.7%.[1,2] There have been numerous reports of risk factors for PEP. According to previous studies, these risk factors have been mentioned as female sex, having a history of acute pancreatitis, precut sphincterotomy, pancreatic duct injection, and so on.[3–5] Some reports of scoring models predict the development of PEP based on these risk factors;[6–8] however, most of these reports were single-center studies and lacked external tests. In addition, they did not evaluate the nonlinear relationships of the factors because the models were developed using statistical methods such as logistic regression (LR) analysis. For the prevention of severe PEP progression, appropriate

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preventive and primary care should be considered, especially for high-risk patients, and the patients should be monitored more closely to receive earlier intensive care when needed. Classifying patients according to their risks is also valuable in terms of effectively using limited medical resources.[9–12] Thus, there is an urgent need for a tool that can predict patients' risk of PEP.

In recent years, remarkable progress has been made with artificial intelligence (AI) technology in a variety of fields, including clinical medicine.[13–15] AI has the ability to greatly surpass human diagnostic imaging[16–18] and prediction of the onset of disease.[19–21] Although there are currently several AI studies using machine-learning algorithms for pancreatitis, these algorithms have only been used to predict acute lung injury,[19] renal failure,[20] and multiorgan failure[21] that follow acute pancreatitis. However, to our knowledge, no studies have been focused on predicting the development of PEP. In this study we aimed to develop an AI-based model to predict patients' risk of PEP.

## METHODS

### Patients

PATIENTS WERE RETROSPECTIVELY identified at Nagoya University Hospital (NUH) and Toyota Memorial Hospital (TMH) in accordance with the following criteria: (i) underwent ERCP for pancreatobiliary disease; and (ii) were older than 20 years of age. Patients were excluded if they met the following criteria: (i) nonnaïve major papilla; (ii) postoperative reconstruction of the upper gastrointestinal tract excluding Billroth-I reconstruction; (iii) cannulation failure; and (iv) ongoing pancreatitis before ERCP. Because of the anonymous nature of the data, the requirement for informed consent was waived. This study was conducted based on the Declaration of Helsinki and approved by the Institutional Review Boards of NUH (No. 2020-0582) and TMH (No. R242), and supported by research grants from The Public Foundation of Chubu Science and Technology Center.

### Definitions of PEP

Post-ERCP pancreatitis was defined as being diagnosed within 24 h after ERCP procedures, and the diagnosis of PEP complied with the revised Atlanta classification.[22] The definition required at least two of the following elements: (i) abdominal pain consistent with acute pancreatitis (acute onset of a persistent, severe, epigastric pain often radiating to the back); (ii) serum lipase activity (or amylase activity) at least three times greater than the upper limit of the standard values; and (iii) characteristic findings of acute pancreatitis on contrast-enhanced computed tomography (CECT), magnetic resonance imaging, or transabdominal ultrasonography. The severity of PEP was diagnosed based on the CECT findings and in compliance with the Japanese guidelines for the management of acute pancreatitis 2015.[23]

### Data collection

We collected data on each patient's demographic characteristics, laboratory examination, primary disease, and therapeutic treatments. We selected 40 possible features, including the risk factors from previous studies,[1–5] for predicting PEP: demographic data (age and sex), body mass index, absence of chronic pancreatitis, history of acute pancreatitis, laboratory examination before ERCP (white blood cell, hemoglobin, platelet, aspartate aminotransferase, alanine aminotransferase, alkaline phosphatase, γ-glutamyl transpeptidase [γGT], total bilirubin, albumin, creatinine, C-reactive protein, carcinoembryonic antigen, and carbohydrate antigen 19–9 levels), primary disease, and procedure-related factors (cannulation time, total procedure time, pancreatic duct injection, pancreatic guidewire-assisted technique [PGW] with or without a pancreatic stent, precut sphincterotomy, pancreatic sphincterotomy, endoscopic sphincterotomy [EST], endoscopic papillary balloon dilatation [EPBD], endoscopic papillary large balloon dilation, intraductal ultrasonography [IDUS], bile duct biopsy, plastic stent across papilla, and metallic stent across papilla). In addition to serum bilirubin, we included common laboratory parameters that are measured in many cases to evaluate the interactions of factors. Patients with ongoing pancreatitis that developed before ERCP were excluded, and pancreatic enzymes were not included as factors because they were within the range of standard values. The cannulation time, total procedure time, and PGW were defined as the time interval from direct visualization of the major papilla to successful cannulation, the time interval from insertion to removal of the endoscope, and the placement of a guidewire into the pancreatic duct to support biliary cannulation,[24] respectively.

### Statistical analysis

The NUH patients were randomly divided into three, roughly equally numbered, nonoverlapping subsets, each called a fold. First, two folds were assigned to the training set, and the other was assigned to the validation set. Then univariate analysis was performed with the training set to analyze 40 features. We performed a three fold cross-validation using the three folds as the validation set and the others as the training set. In addition, the models were tested with TMH patients.

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Continuous data are presented as medians with interquartile ranges or means ± standard deviations. Categorical data are presented as frequencies with proportions. Between-group differences were assessed using a two-sided *t*-test, Mann–Whitney *U*-test, or Fisher's exact test, as appropriate. The regression analysis results are presented in terms of the estimated odds ratios with corresponding 95% confidence intervals (CIs). *P*-values of <0.05 were considered statistically significant. The area under the receiver operating characteristic curve (AUROC) was used to assess the performance of the prediction models.

## Model development

We constructed a random forest (RF), one of the machine-learning algorithms, and an LR model to predict PEP using the clinical parameters of the patients. A model based on the RF classifier was constructed using bootstrap sampling, and an RF provided a test set consisting of 37% observations withheld to assess the predictive performance and avoid overestimating.[25] The out-of-bag classification error was also calculated, which involves an error estimation technique often used to evaluate model accuracy and select appropriate values for tuning parameters. To develop the RF model, we first used the 40 features as the input, and then the wrapper method was used to select the features to increase the accuracy of the RF models. The wrapper method was used to select features from a feature set and evaluate the AUROC value. If the AUROC values converged, we terminated this iteration process. The RF model consisted of 100 trees. The remaining parameters of the RF model were set to default as implemented in the MATLAB (MathWorks, Natick, MA, USA) machine-learning package. After the feature selection process, we performed a grid search to find the model parameter that achieved a higher AUROC for the training set. To develop the LR model, features with a *P*-value <0.10 in the univariate analysis were entered into the multivariate LR analysis using the forward stepwise selection method. In the forward stepwise selection method, features are added one-by-one based on their contribution to the model's performance. Considering clinical importance, these models converted the laboratory data to categorical features, categorized as within or outside the range of standard values. Furthermore, we classified the patients into low-, medium-, and high-risk groups using the posterior probabilities of the RF model. The tertile posterior probabilities of the training set were defined as the cut-off and applied to the validation set and test set.

All statistical analyses and machine-learning modeling were performed using R version 4.1.0 (R Foundation for Statistical Computing, Vienna, Austria), IBM SPSS Statistics v. 27 (IBM, Armonk, NY, USA), and MATLAB 2021a.

# RESULTS

## Patient characteristics

We retrospectively reviewed 2065 patients who underwent ERCP at NUH and 981 patients who underwent ERCP at TMH from January 2018 to December 2020. After excluding the patients with nonnaïve major papilla, postoperative reconstruction of the upper gastrointestinal tract excluding Billroth-I reconstruction, cannulation failure, and ongoing pancreatitis before ERCP, a total of 615 patients at NUH and 544 patients at TMH were enrolled in this study (Fig. 1). NUH is an academic center, while TMH is a regional core hospital. Therefore, the backgrounds of the patients were quite different (Table 1). At NUH, the ERCP procedures were mainly performed for patients with malignancy, but in TMH, they were mainly performed for patients with benign disease. There were no significant differences in the baseline characteristics, except for patient age. The proportions of patients with PEP in NUH and TMH were 10.2% and 9.6%, respectively. In addition, the proportions of patients with severe PEP were 1.1% and 2.6%, respectively. There was no significant difference in the incidence of overall PEP or severe PEP between the two hospitals.

## Univariate analysis

The results of the univariate analysis in the training set with the 40 features are shown in Table 2. In the PEP group, the platelet level ($P = 0.023$), proportion of patients with biliary tract cancer ($P < 0.001$), PGW without a pancreatic stent ($P = 0.015$), and bile duct biopsy ($P = 0.002$) were significantly higher than those in the non-PEP group.

## RF model

The wrapper method was utilized to develop the RF model (Fig. 2). After adding features to maximize the AUROC of the training set, it peaked with 21 features. However, as the number of features increased, the out-of-bag classification error also increased, suggesting overfitting. Therefore, we developed the model with 10 features where the out-of-bag classification error was the lowest. The features included albumin, creatinine, biliary tract cancer, pancreatic cancer, bile duct stone, total procedure time >30 min, pancreatic duct injection, PGW without a pancreatic stent, IDUS, and bile duct biopsy. The grid search selected the model parameters (InBagFraction: 0.5, Cost: [0 1; 20 0], and MinLeafSize: 2 in MATLAB machine-learning package). The feature importance of the model was calculated in the process (Fig. 3). In the three fold cross-

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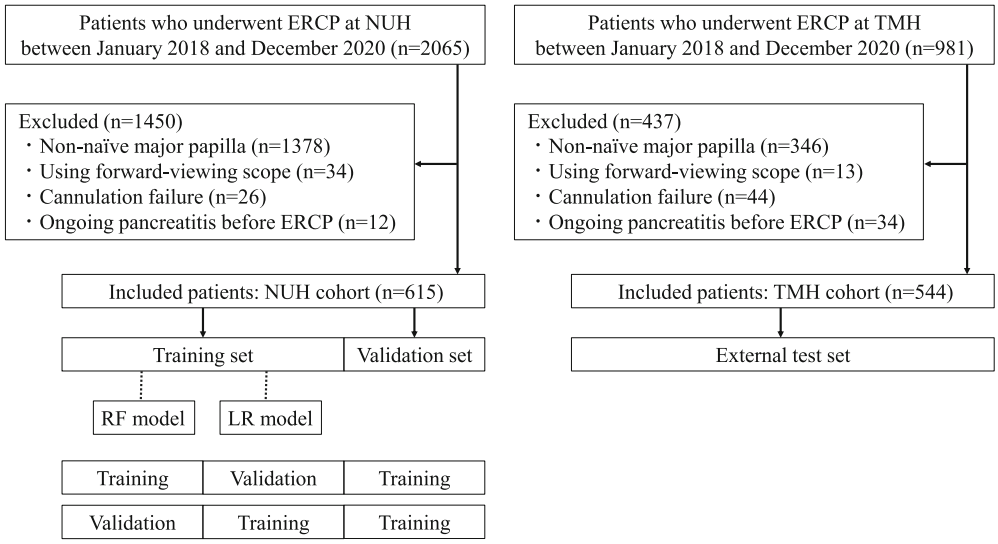



**Figure 1** Patient flow. The patients at Nagoya University Hospital (NUH, *n* = 615) were assigned to the training and validation sets, and those at Toyota Memorial Hospital (TMH, *n* = 544) were assigned to the external test set. Internal three fold cross-validation was performed with the NUH cohort. Then the external test was performed with the TMH cohort. ERCP, endoscopic retrograde cholangiopancreatography; LR, logistic regression; RF, random forest.

**Table 1** Characteristics of the patients at Nagoya University Hospital (NUH) and Toyota Memorial Hospital (TMH)

| Features | NUH (*n* = 615) | TMH (*n* = 544) | *P*-value |
|---|---|---|---|
| Baseline characteristics | | | |
| Age, years, median (IQR) | 70.0 (61.0–77.0) | 74.0 (64.0–81.0) | **<0.001** |
| Age <60, *n* (%) | 143 (23.3) | 114 (21.0) | 0.358 |
| Age <40, *n* (%) | 27 (4.4) | 17 (3.1) | 0.284 |
| Female, *n* (%) | 228 (37.1) | 224 (41.2) | 0.165 |
| BMI, kg/m$^2$, median (IQR) | 22.2 (20.2–24.7) | 21.9 (19.3–24.9) | 0.160 |
| PEP, *n* (%) | 63 (10.2) | 52 (9.6) | 0.768 |
| Severe PEP, *n* (%) | 7 (1.1) | 14 (2.6) | 0.079 |
| Primary disease, *n* (%) | | | |
| Biliary tract cancer | 136 (22.1) | 48 (8.8) | **<0.001** |
| Pancreatic cancer | 107 (17.4) | 41 (7.5) | **<0.001** |
| Other malignancy or precancerous lesion | 106 (17.2) | 31 (5.7) | **<0.001** |
| Bile duct stone | 170 (27.6) | 411 (75.6) | **<0.001** |
| Pancreaticobiliary malformation | 19 (3.1) | 3 (0.6) | **0.002** |
| Other benign disease | 77 (12.5) | 10 (1.8) | **<0.001** |

Bold values indicate significance.
BMI, body mass index; IQR, interquartile range; PEP, postendoscopic retrograde cholangiopancreatography pancreatitis.

validation, the average AUROC was 0.821 (fold1: 0.815, fold2: 0.795, fold3: 0.853). In the external test with the TMH cohort, the AUROC was 0.770 (Fig. 4).

## LR model

The eight features that showed *P*-values less than 0.10 in the univariate analysis were used for the LR analysis (platelets, γGT, biliary tract cancer, bile duct stone, PGW without a pancreatic stent, EST, EPBD, and bile duct biopsy). The forward stepwise method was utilized, and four features were selected for the model (biliary tract cancer, PGW without a pancreatic stent, EST, and EPBD) (Table 3). In the three fold cross-validation, the average AUROC was 0.660 (fold1: 0.678, fold2: 0.561, fold3: 0.741). In the external test, the average AUROC was 0.663 (Fig. 4). Comparing the

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**Table 2** Univariate analysis of the training set

| Features | Standard value | PEP group ($n$ = 47) | Non-PEP group ($n$ = 366) | Odds ratio (95% CI) | $P$-value |
|---|---|---|---|---|---|
| Patient-related factors | | | | | |
| Age, years, median (IQR)[‡] | – | 71.0 (62.0–77.0) | 69.0 (61.0–76.0) | – | 0.460 |
| Age <60, $n$ (%) | – | 11 (23.4) | 85 (23.2) | 1.01 (0.49–2.07) | 1.000 |
| Age <40, $n$ (%)[†] | – | 0 (0.0) | 18 (4.9) | N.A | 0.244 |
| Female, $n$ (%)[†,‡] | – | 21 (44.7) | 141 (38.5) | 1.29 (0.70–2.38) | 0.431 |
| BMI, kg/m$^2$, mean ± SD | – | 22.8 ± 3.2 | 22.6 ± 3.6 | – | 0.735 |
| Absence of chronic pancreatitis, $n$ (%)[†,‡] | – | 46 (97.9) | 359 (98.1) | 0.90 (0.11–7.46) | 0.897 |
| Previous acute pancreatitis, $n$ (%)[†,‡] | – | 1 (2.1) | 9 (2.5) | 0.86 (0.11–6.96) | 0.862 |
| Laboratory findings, median (IQR) | | | | | |
| WBC, /μL | 3300–8600 | 5900 (5225–6875) | 5800 (4700–7300) | – | 0.449 |
| Hb, g/dL | 11.6–14.8 | 13.2 (11.5–14.1) | 12.7 (11.4–14.0) | – | 0.372 |
| Platelet, /μL | 158,000–348,000 | 248,000 (199,250–289,750) | 218,000 (180,000–270,000) | – | **0.023** |
| AST, U/L | 13–30 | 39 (20–164) | 44 (21–123) | – | 0.934 |
| ALT, U/L | 7–23 | 53 (18–195) | 51 (19–167) | – | 0.730 |
| Alkaline phosphatase, U/L | 106–322 | 582 (206–1001) | 442 (234–840) | – | 0.555 |
| γGT, U/L | 9–32 | 302 (57–692) | 187 (29–488) | – | 0.083 |
| Total bilirubin, mg/dL[†,‡] | 0.4–1.5 | 1.2 (0.7–4.3) | 1.0 (0.7–2.3) | – | 0.246 |
| Albumin, g/dL | 4.1–5.1 | 3.9 (3.5–4.2) | 3.8 (3.3–4.2) | – | 0.363 |
| Creatinine, mg/dL | 0.46–0.79 | 0.76 (0.62–0.89) | 0.78 (0.63–0.95) | – | 0.480 |
| CRP, mg/dL | −0.14 | 0.31 (0.06–1.05) | 0.23 (0.07–1.47) | – | 0.610 |
| CEA >5 ng/mL, $n$ (%) | – | 11 (23.4) | 62 (16.9) | 1.50 (0.72–3.10) | 0.309 |
| CA19-9 >37 U/mL, $n$ (%) | – | 20 (42.6) | 125 (34.2) | 1.43 (0.77–2.65) | 0.260 |
| Diagnosis of the patients, $n$ (%) | | | | | |
| Biliary tract cancer | – | 19 (40.4) | 67 (18.3) | 3.03 (1.60–5.74) | **0.001** |
| Pancreatic cancer | – | 7 (14.9) | 67 (18.3) | 0.78 (0.34–1.82) | 0.688 |
| Other malignancy or precancerous lesion | – | 6 (14.3) | 70 (19.1) | 0.62 (0.25–1.51) | 0.423 |
| Bile duct stone | – | 7 (14.9) | 102 (27.9) | 0.45 (0.20–1.04) | 0.077 |
| Pancreaticobiliary malformation | – | 3 (6.4) | 11 (3.0) | 2.20 (0.59–8.19) | 0.206 |
| Other benign disease | – | 5 (10.6) | 49 (13.4) | 0.77 (0.29–2.04) | 0.818 |
| Procedure-related factors, $n$ (%) | | | | | |
| Cannulation time >10 min[†] | – | 15 (31.9) | 85 (23.2) | 1.55 (0.80–3.00) | 0.207 |
| Total procedure time >30 min | – | 5 (10.6) | 17 (4.6) | 2.44 (0.86–6.96) | 0.091 |
| Pancreatic duct injection[†,‡] | – | 32 (68.1) | 210 (57.4) | 1.58 (0.83–3.03) | 0.208 |
| PGW with a pancreatic stent | – | 3 (6.4) | 21 (5.7) | 1.12 (0.32–3.91) | 0.745 |
| PGW without a pancreatic stent | – | 13 (27.7) | 48 (13.1) | 2.53 (1.25–5.14) | **0.015** |
| Precut sphincterotomy[‡] | – | 0 (0.0) | 4 (1.1) | N.A | 1.000 |
| Pancreatic sphincterotomy[†,‡] | – | 1 (2.1) | 6 (1.6) | 1.30 (0.15–11.08) | 0.574 |
| EST | – | 7 (14.9) | 106 (29.0) | 0.43 (0.19–0.99) | 0.055 |
| EPBD[‡] | – | 4 (8.5) | 10 (2.7) | 3.31 (1.00–11.02) | 0.063 |
| EPLBD[†] | – | 1 (2.1) | 7 (1.9) | 1.11 (0.13–9.27) | 1.000 |
| IDUS | – | 26 (55.3) | 163 (44.5) | 1.54 (0.84–2.84) | 0.166 |
| Bile duct biopsy | – | 21 (44.7) | 80 (21.9) | 2.89 (1.54–5.40) | **0.002** |

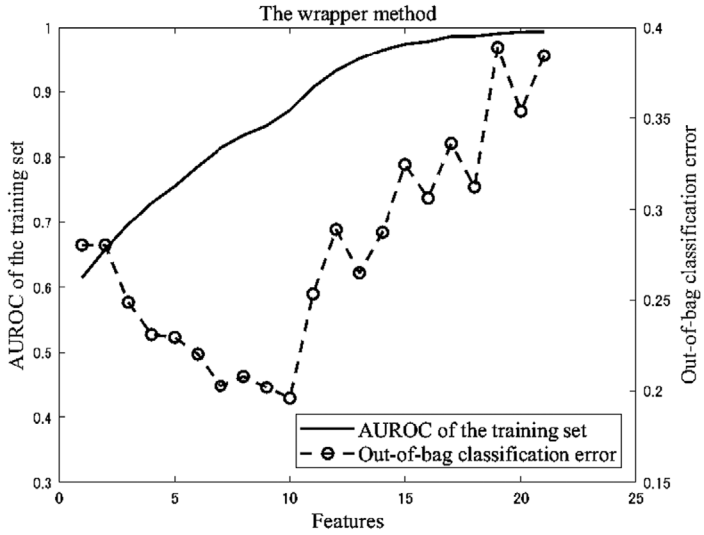
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**Table 2** (*Continued*)

| Features | Standard value | PEP group ($n$ = 47) | Non-PEP group ($n$ = 366) | Odds ratio (95% CI) | $P$-value |
|---|---|---|---|---|---|
| Plastic stent across papilla | – | 4 (8.5) | 42 (11.5) | 0.72 (0.25–2.10) | 0.805 |
| Metallic stent across papilla | – | 7 (14.9) | 44 (12.0) | 1.28 (0.54–3.03) | 0.636 |

†Twelve risk factors from the 2017 American Society for Gastrointestinal Endoscopy guidelines for PEP.
‡Seven definitive risk factors or 10 likely risk factors from the 2020 European Society of Gastrointestinal Endoscopy guidelines for PEP.
Bold values indicate signficance.
ALT, alanine aminotransferase; AST, aspartate aminotransferase; BMI, body mass index; CA19-9, carbohydrate antigen 19-9; CEA, carcinoembryonic antigen; CI, confidence interval; CRP, C-reactive protein; EPBD, endoscopic papillary balloon dilatation; EPLBD, endoscopic papillary large balloon dilation; EST, endoscopic sphincterotomy; Hb, hemoglobin; IDUS, intraductal ultrasonography; IQR, interquartile range; N.A, not applicable; PEP, postendoscopic retrograde cholangiopancreatography pancreatitis; PGW, pancreatic guidewire-assisted technique; SD, standard deviation; WBC, white blood cell; γGT, γ-glutamyl transpeptidase.




| | Features | AUROC | OOB error |
|---|---|---|---|
| 1 | Bile duct biopsy | 0.614 | 0.280 |
| 2 | PGW without pancreatic stent | 0.658 | 0.280 |
| 3 | IDUS | 0.697 | 0.249 |
| 4 | Bile duct stone | 0.730 | 0.231 |
| 5 | Pancreatic injection | 0.756 | 0.230 |
| 6 | Creatinine | 0.786 | 0.221 |
| 7 | Pancreatic cancer | 0.814 | 0.203 |
| 8 | Albumin | 0.834 | 0.208 |
| 9 | Biliary tract cancer | 0.848 | 0.202 |
| 10 | Total procedure time >30 minutes | 0.871 | 0.196 |

**Figure 2** The wrapper method. The wrapper method was performed to select the features for the random forest model. AUROC, area under the receiver operating characteristic curve; IDUS, intraductal ultrasonography; OOB, out-of-bag; PGW, pancreatic guidewire-assisted technique.

two models, the RF model showed significantly better performance than the LR model ($P$ = 0.002).

## Prediction of patient risk

We used the posterior probabilities of the RF model to classify the patients into three groups. In the validation set, the incidence of PEP was 0.0% in the low-risk group, 4.5% in the medium-risk group, and 28.9% in the high-risk group. In the external test set, the incidence of PEP was 2.9% in the low-risk group (7 of 245 patients), 10.0% in the medium-risk group (19 of 190 patients), and 23.9% in the high-risk group (26 of 109 patients). The patients were significantly classified as at risk of PEP ($P$ < 0.05, Fig. 5).

## DISCUSSION

IN THIS STUDY we developed novel prediction models for PEP. In the three fold cross-validation and external test, the RF model showed significantly better performance than the LR model ($P$ = 0.002, Fig. 4). In addition, the posterior probabilities of the RF model were useful to classify the patients into low-, medium-, and high-risk groups (2.9%, 10.0%, and 23.9%, respectively, Fig. 5).

Previously, no scoring system was widely accepted for the development of PEP; thus, a reference model was needed to assess the performance of the models. LR analysis is a useful method that has been widely used in many previous studies.[6–8,19–21] However, in this study the RF model

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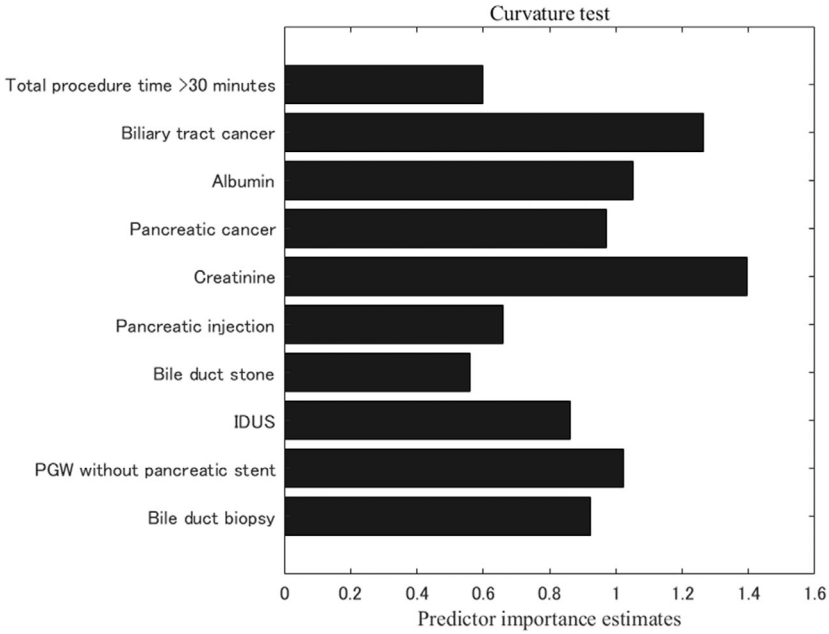



**Figure 3** Feature importance. The feature importance of the random forest model was calculated. IDUS, intraductal ultrasonography; PGW, pancreatic guidewire-assisted technique.

showed better performance than the LR model. This finding was supportive of the RF classifier being a useful method to predict the risk of PEP. Multiple factors play complex roles in the development of PEP;[26,27] thus, it is essential to evaluate the nonlinear interactions of the factors. Furthermore, in the wrapper method, the features of the RF model were selected to maximize the AUROC of the training set and to minimize the out-of-bag classification error (Fig. 2). In this process, we selected 10 important features out of 40 possible features. This contributed to the development of the generalizable model.

In the univariate analysis, more patients in the PEP group had biliary tract cancer ($P < 0.001$). Regarding procedure-related factors, PGW without a pancreatic stent and bile duct biopsy showed a significant risk for PEP ($P = 0.015$ and $P = 0.002$, respectively, Table 2). In contrast, PGW with a pancreatic stent showed no significant risk, suggesting that the stent may work in a preventive way for PEP, as previously reported.[28–30] In the selection of features using the stepwise selection method, four features were selected for the LR model (biliary tract cancer, PGW without a pancreatic stent, EST, and EPBD) (Table 3). Using the wrapper method, 10 features were selected for the RF model (albumin, creatinine, biliary tract cancer, pancreatic cancer, bile duct stone, total procedure time >30 min, pancreatic duct injection, PGW without a pancreatic stent, IDUS, and bile duct biopsy) (Fig. 2). The feature importance of the RF model showed that creatinine had the largest effect on the prediction (Fig. 3), which suggested that renal disease was a risk factor for PEP, as previously reported.[31,32]

This study has the following strengths. We used patient data from two hospitals. The backgrounds of the patients at each hospital were quite different, especially regarding the disease rates (Table 1). Nevertheless, the model showed good performance in the external test. Moreover, the RF model could classify the patients into low-, medium-, and high-risk groups. ERCP accommodates various patients and procedures; thus, patients should be treated based on their own risk. By incorporating an accurate prediction model into clinical practice, clinicians can effectively identify high-risk patients. This facilitates the delivery of appropriate primary care, enables earlier initiation of intensive care, and optimizes the allocation of medical resources. In addition to careful patient selection and informed decision-making, the model allows for strong recommendations for pancreatic stenting and rectal nonsteroidal antiinflammatory drugs, based on evidence for PEP prevention.[33–36] Furthermore, we can be more timely in resuming meal schedules and initiating rehabilitation after the procedures for low-risk patients and reduce unnecessary admissions.

There are several limitations in this study. Some risk factors from the international guidelines showed no

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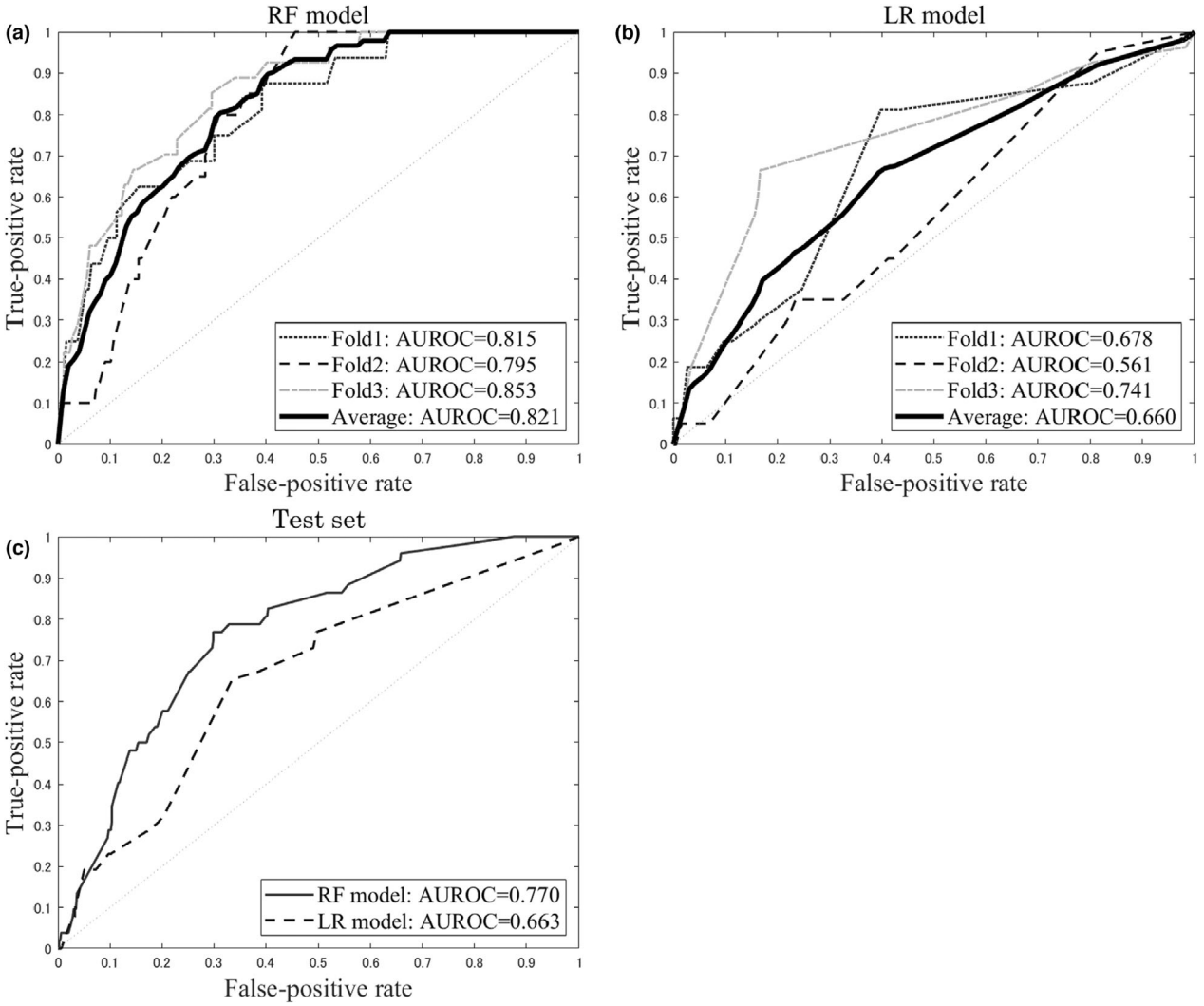



**Figure 4** The model performance. Three fold cross-validation was performed for the random forest (RF) and logistic regression (LR) models with the Nagoya University Hospital cohort. (a) The area under the receiver operating characteristic curve (AUROC) of the RF model: fold1 = 0.815, fold2 = 0.795, fold3 = 0.853, average AUROC = 0.821. (b) The AUROC of the LR model: fold1 = 0.678, fold2 = 0.561, fold3 = 0.741, average AUROC = 0.660. (c) The external test was performed with the Toyota Memorial Hospital cohort. The AUROC of the RF model was significantly larger than that of the LR model (0.770, 0.663, $P = 0.002$).

significance in the statistical analysis (Table 2). It was possible that they were not evaluated properly due to an insufficient number of cases. There were exclusion criteria for the patients; thus, the models were not available for all ERCP patients (Fig. 1). In this study, only patients who completed the intended ERCP-related procedure were included; thus, patients with cannulation failure were excluded. In addition, we excluded patients with postoperative reconstruction of the upper gastrointestinal tract, excluding Billroth-I reconstruction. In the postoperative patients, we used forward-viewing scopes, including balloon-assisted endoscopes, because of the gastrointestinal anatomy; therefore, the approaches of ERCP procedures are considered different. In addition to the LR analysis and RF classifier, there are many other algorithms that can be adapted to classification tasks, such as the k-nearest neighbor algorithm, support vector machine, and neural network. Machine-learning algorithms have great potential to evaluate more complex factor relationships. However, the more complex the model is, the more difficult it may become to understand why the result was derived. Due

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**Table 3** Stepwise method (logistic regression analysis)

| Features | Partial regression coefficient | *P*-value | Odds ratio (95% CI) |
|---|---|---|---|
| Biliary tract cancer | 0.984 | **0.004** | 2.674 (1.361–5.253) |
| Pancreatic guidewire-assisted technique without pancreatic stent | 1.279 | **0.002** | 3.594 (1.601–8.071) |
| Endoscopic sphincterotomy | −1.003 | **0.039** | 0.367 (0.141–0.953) |
| Endoscopic papillary balloon dilatation | 1.226 | 0.058 | 3.407 (0.958–12.116) |
| Constant | −2.438 | 0.000 | |

Bold values indicate significance.
CI, confidence interval.

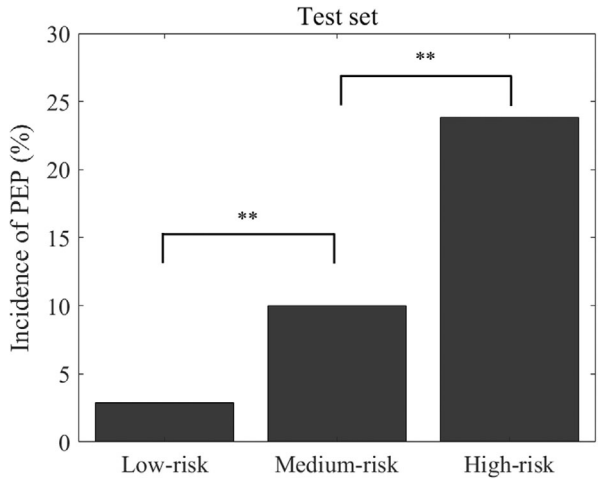



**Figure 5** Classification of the patients. The patients were classified into low-, medium-, and high-risk groups according to the posterior probabilities of the random forest model. In the validation set, the incidence rates of postendoscopic retrograde cholangiopancreatography pancreatitis (PEP) were 0.0%, 4.5%, and 28.9%. In the external test set, the incidence rates of PEP were 2.9%, 10.0%, and 23.9%. $^{**}P < 0.05$.

to these limitations, further investigations using multicenter studies with larger sample sizes are needed. In addition, medical professionals should understand the characteristics of machine-learning algorithms and make appropriate use of them to achieve the best clinical performance.[37–40]

In conclusion, we developed prediction models for PEP. The RF model showed better performance than the LR model and was able to significantly classify the patients' risk of PEP. Machine-learning algorithms could be powerful tools to develop accurate prediction models of PEP that involve multiple contributing factors. With the extension of case numbers and the construction of a more generalized prediction model, the implementation of such models as applications becomes feasible, enabling personalized care for patients based on their individual risks.

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## CONFLICT OF INTEREST

AUTHORS DECLARE NO conflict of interest for this article.

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